

9.2 Runlevels

Runlevel is a preset state for Linux operating system. A system can be booted into any of several runlevels, each of which is represented by a single digit integer. Each runlevel designates a different system configuration and allows access to a different combination of processes. For example, runlevel 3 does not start the GUI services.

Seven runlevels are supported in the standard Linux kernel. They are:

- 0 — System halt; no activity, the system can be safely powered down.
- 1 — Single user mode.
- 2 — Multiple users, no NFS (network filesystem); (user defined).
- 3 — Multiple users, command-line (i.e., all-text mode) interface; the standard runlevel for most Linux-based server machines.
- 4 — User-definable
- 5 — Multiple users, X11 (GUI); it is the standard for Desktop-level Linux machines.
- 6 — Reboot; used when restarting the system from Command Line Interface (CLI).

By default Linux boots either to runlevel 3 or to runlevel 5. The former permits the system to run all services except for a GUI. The latter allows all services including a GUI.

The program responsible for changing runlevel or booting a pre-defined runlevel is *init*. To change from one runlevel to another the '*telinit <runlevel>*' command can be issued.

Eg:

#telinit 3 + [Enter] to change from runlevel 5 to 3. Assuming that current runlevel is 5.

As a system administrator, let's get down to the depth of runlevels. A file */etc/inittab* has an entry for the default runlevel. In order to manually change the default runlevel edit the line:

id:5:initdefault: